

SEE MODEL QUESTION 2080 (ISSUED BY CDC)

Science and Technology — Complete Solved Paper (All Questions Included)
English Medium • Textbook Level Explanations • Spacious Format

Group 'A' — Multiple Choice Questions (10 x 1 = 10 Marks)

(a) Which is the first step at the signal receiver side?

Correct Answer: (iii) Demodulator

The causes of this to happen are: The incoming signal travels over a high-frequency carrier wave. When the signal reaches the receiving side, the very first tool it passes through is the demodulator to separate the original message back out from the carrier wave.

Simple explanation: Think of a text message being carried inside a postman's bag. When it arrives at your house (the receiver), the first step is to take the letter out of the bag. This unpacking step is called demodulation.

(b) Why is cycas kept in gymnosperm?

Correct Answer: (iv) It has cone instead of flower and seed is naked.

The causes of this to happen are: Cycas is a primitive plant that does not produce true biological flowers or real fruits. Instead of flowers, it grows woody reproductive parts called cones, and its seeds sit completely bare and exposed ('naked') directly on these cones.

Simple explanation: Normal plants hide their seeds inside protective fruits like apples or mangoes. Gymnosperms do not grow flowers or fruits at all, so their seeds are open and completely naked to the air.

(c) Organism A has a cylindrical segmented body and organism B has a non-segmented body. Which statement is true for these organisms?

Correct Answer: (i) Organism A is hermaphrodite whereas organism B is unisexual.

The causes of this to happen are: Organism A has rings on its body, which means it belongs to the earthworm family (Annelida) where each single individual contains both male and female systems (hermaphrodite). Organism B has a smooth body, meaning it belongs to the roundworm family (Nematoda) where males and females are completely separate individuals (unisexual).

Simple explanation: Organism A is an earthworm, which has a body made of rings. It has both male and female parts inside one body. Organism B is a smooth roundworm, which has separate male and female worms.

(d) Yarsagumba is formed by the fusion of caterpillar and fungi. What is the interrelation between caterpillar and fungus in this process?

Correct Answer: (ii) Parasitic

The causes of this to happen are: The fungus behaves as a harmful guest that invades a living ghost moth caterpillar. The fungus stays inside, eats the caterpillar's internal body tissues for food, and completely kills the caterpillar so it can grow its mushroom stem out of the insect's head.

Simple explanation: In a parasitic relationship, one side steals all the food and life while the other side gets hurt and dies. The fungus eats the caterpillar from the inside out.

(e) The mass of the earth is 6×10^{24} kg. What would be the acceleration due to gravity of the meteor falling from 10000 km height from the centre of the earth?

Correct Answer: (ii) 4.002 m/s^2

The full working process using textbook equations is shown below:

Given parameters:

- Mass of Earth (M) = 6×10^{24} kg
- Distance from Earth center (R) = $10000 \text{ km} = 10000 \times 1000 \text{ m} = 10^7 \text{ m}$
- Gravitational Constant (G) = $6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$

Formula:

$$g = (G \times M) / R^2$$

Calculation:

$$g = (6.67 \times 10^{-11} \times 6 \times 10^{24}) / (10^7)^2$$

$$g = (40.02 \times 10^{13}) / 10^{14}$$

$$g = 40.02 \times 10^{-1} = 4.002 \text{ m/s}^2$$

Simple explanation: Gravity gets weaker when you move away from Earth. On the surface, gravity is 9.8 m/s^2 . Out in space at 10,000 km from the center, the gravity drop leaves it at exactly 4.002 m/s^2 .

(f) Why do objects float more easily in sea water than in river?

Correct Answer: (i) Density of sea water is more than that of river.

The causes of this to happen are: Sea water is full of dissolved table salt, which makes it heavy and dense. According to our textbook, higher liquid density creates a much bigger upward push called upthrust, which supports floating objects easily.

Simple explanation: Salt makes sea water thick and heavy. This heavy water pushes up tightly on anything dropped into it, making floating very easy compared to fresh river water.

(g) Sarthak named red, green, violet, yellow and blue coloured ray in dispersion of speed in prism as v_r , v_g , v_v , v_y and v_b respectively. Which order is correct?

Correct Answer: (iii) $v_g < v_y < v_r > v_v$

The causes of this to happen are: Inside a glass prism, red light travels fastest because glass slows it down the least (v_r is maximum). Violet light travels slowest because glass delays it the most (v_v is minimum). Therefore, red is faster than yellow, which is faster than green, which is faster than blue, which is faster than violet.

Simple explanation: Red light runs the fastest inside glass, while violet light is the slowest runner. This option correctly shows red at the highest speed peak.

(h) Which is the most realistic theory about origin of the universe?

Correct Answer: (iv) Big Bang theory

The causes of this to happen are: Scientists have clear lab proof that all galaxies are moving far apart from each other, which proves that the entire universe is expanding. The Big Bang theory correctly explains that everything started from a tiny, hot single point that burst outward billions of years ago.

Simple explanation: The Big Bang is the best theory because it perfectly explains why space is expanding today.

(i) What is denoted by X in the given reaction? $2\text{KClO}_3 \rightarrow (\text{with X}) \rightarrow 2\text{KCl} + 3\text{O}_2$

Correct Answer: (i) Catalyst

The causes of this to happen are: In the laboratory setup for oxygen gas, Manganese dioxide is written over the arrow as 'X'. It acts as a positive helper that increases the speed of chemical breaking without changing itself. This helper is called a catalyst.

Simple explanation: A catalyst is a chemical helper that speeds up a slow reaction without getting used up.

(j) Statement: Stomach pesticide is a type of pesticide based on mode of action. Argument 1: Insects die when they eat leaves and fruits treated by pesticides. Argument 2: When these pesticides come contact with air, poisonous gas is released.

Correct Answer: (ii) Statement and argument 1 is correct but argument 2 is incorrect

The causes of this to happen are: A stomach poison can only work if an insect physically chews and swallows the food into its belly (Argument 1). Argument 2 is wrong because a pesticide that turns into gas is called a fumigant, which is a different category.

Simple explanation: A stomach poison requires the insect to eat the crop so the medicine goes into its stomach. It does not release dangerous gas into the air.

Group 'B' — Very Short Answer Questions (9 x 1 = 9 Marks)

(a) Why are controlled variables important?

The causes of controlled variables being important are: They keep all background test factors exactly the same so that the experiment remains a fair test, ensuring that the results are only caused by the one variable we want to test.

(b) State Archimedes' principle.

The fundamental statement of Archimedes' principle is: When an object is partially or completely dipped into a liquid, it experiences an upward push called upthrust which is exactly equal to the weight of the liquid pushed away by the object.

(c) Anu cannot distinguish red and green colour. What is the main cause of this problem?

The main cause of this problem is: A lack or a defect in the color-sensitive cone cells located inside the retina layer at the back of her eyes.

(d) A student of the class can see letters written on instructional board when s/he sits on first bench and cannot see clearly when s/he sits on last bench. What type of lens is used to remove this defect?

The type of lens used to remove this visual defect is a: Concave lens (because the student has short-sightedness, meaning they can see near things but faraway things look blurry).

(e) What happens when a bar magnet is placed inside a solenoid at rest?

The result of placing a bar magnet inside a solenoid at rest is: Absolutely nothing happens, and the current meter stays at zero because there is no relative movement to cut the magnetic lines.

(f) What is the main cause of determination of the future of the universe?

The main cause of determination of the future of the universe is: The average density of matter that exists across space.

(g) Write the name of the main ore of iron.

The name of the main commercial ore used to extract iron is: Hematite.

(h) Pratik connected impure copper to the negative terminal of a cell and pure copper to the positive terminal of the cell and copper sulphate is used as electrolyte. What is the mistake he made in this process?

The mistake he made in this process is: He connected the impure copper to the negative terminal and pure copper to the positive terminal, whereas he should have connected the impure copper to the positive terminal (anode) and the pure copper to the negative terminal (cathode).

(i) Write a difference between the given hydrocarbons on the basis of reactivity.

The primary difference on the basis of reactivity is: Figure A (Ethene) is highly reactive because it contains a double chemical bond, while Figure B (Ethane) is less reactive because it contains only stable single bonds.

Group 'C' — Short Answer Questions (14 x 2 = 28 Marks)**3. Prove that unit of electric resistance (Ohm) = $\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3} \cdot \text{A}^{-2}$.**

The step-by-step verification with full equations is shown below:

By Ohm's Law, Resistance (R) = Voltage (V) / Current (I)

We know Voltage (V) = Work Done (W) / Charge (Q)

And Charge (Q) = Current (I) × Time (t)

So, Voltage (V) = W / (I × t)

Substitute V into the Resistance formula:

$$R = W / (I^2 \times t)$$

We know Work Done (W) = Force (F) × Distance (d)

And Force (F) = Mass (m) × Acceleration (a)

So, Work Done (W) = Mass (m) × Acceleration (a) × Distance (d)

Now, replace everything into the unit equation:

$$\text{Unit of } R = (\text{kg} \times \text{m/s}^2 \times \text{m}) / (\text{A}^2 \times \text{s})$$

$$\text{Unit of } R = (\text{kg} \times \text{m}^2 \times \text{s}^{-2}) / (\text{A}^2 \times \text{s}^1)$$

$$\text{Unit of } R = \text{kg} \times \text{m}^2 \times \text{s}^{-2-1} \times \text{A}^{-2}$$

$$\text{Ohm} = \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3} \cdot \text{A}^{-2} \text{ — Hence Proved.}$$

4. Write any two characteristics of the class amphibia.

Two simple characteristics from our textbook are:

1. They can live comfortably both on dry land and inside water.
2. They are cold-blooded animals with a moist, slippery skin that does not have any scales.

5. Study the given concept map and name A and B.

The names of the missing honey bee categories are:

- **A: Female Bee (Worker Bee or Queen Bee)** — Developed from a regular fertilized egg.
- **B: Male Bee (Drone Bee)** — Developed directly from an unfertilized egg through parthenogenesis.

6. Write any two differences between somatic chromosome and sex chromosome.

Two textbook differences are:

1. Somatic chromosomes determine general body traits like height and skin color, whereas sex chromosomes determine whether the body is male or female.
2. In human cells, there are 22 pairs of somatic chromosomes, but there is only 1 pair of sex chromosomes.

7. Draw a chart of the result obtained by crossing pure black mouse (BB) and hybrid black mouse (Bb).

The complete genetic cross layout is given below:

Parents: Pure Black (BB) × Hybrid Black (Bb)

Gametes: (B, B) from pure parent and (B, b) from hybrid parent.

Punnett Square Chart:

	B	b
B	BB (Pure Black)	Bb (Hybrid Black)
B	BB (Pure Black)	Bb (Hybrid Black)

Result: 50% Pure Black mice (BB) and 50% Hybrid Black mice (Bb). All offspring are phenotypically black.

8. White blood cell of Renu's blood is found to be 2500 after examining her blood. What are the possible causes of this? Write any two causes.

The causes of this to happen are: Renu's WBC count is below the normal range (4000 to 11000). Two simple textbook reasons for this drop are:

1. A recent viral infection that temporarily slowed down the cell production inside her bone marrow.
2. An autoimmune condition where the body accidentally destroys its own protective white blood cells.

9. Despite the medicinal herbs in our locality, we are dependent upon imported medicines. Write its one cause and what can we do to increase the use of medicinal herbs? Write any one point.

One main cause of dependency: We lack local modern processing factories and standard dosage formulas for our wild herbs.

One action point to increase use: We should set up small extraction labs locally to pack pure herbs into simple, clean tablets and capsules.

10. Write any two effects of gravity.

Two simple textbook effects of gravity are:

1. It causes an object dropped from a height to speed up as it falls down toward the earth.
2. It holds the layer of breathing air (atmosphere) tight around the surface of our planet.

11. Kabin dropped a stone from a tower of 49 m high. One second later, he threw another stone. They both hit the ground at the same time. Find out the velocity with which he threw the second stone.

The full mathematical process with equations is shown below:

Step 1: Motion of the First Stone (Dropped from rest)

• Height (h) = 49 m, Initial speed (u_1) = 0 m/s, Gravity (g) = 9.8 m/s^2

$$\text{Equation: } h = u_1 \cdot t_1 + 0.5 \cdot g \cdot t_1^2$$

$$49 = (0 \times t_1) + (0.5 \times 9.8 \times t_1^2)$$

$$49 = 4.9 \times t_1^2$$

$$t_1^2 = 49 / 4.9 = 10$$

$$t_1 = \sqrt{10} = 3.16 \text{ seconds}$$

Step 2: Motion of the Second Stone (Thrown downward 1 second later)

• It leaves 1 second later but lands at the same time, so its time is: $t_2 = 3.16 - 1 = 2.16$ seconds

• Height (h) = 49 m, Initial speed = u_2

$$\text{Equation: } h = u_2 \cdot t_2 + 0.5 \cdot g \cdot t_2^2$$

$$49 = u_2 \times (2.16) + (0.5 \times 9.8 \times (2.16)^2)$$

$$49 = 2.16 \times u_2 + (4.9 \times 4.6656)$$

$$49 = 2.16 \times u_2 + 22.86$$

$$2.16 \times u_2 = 49 - 22.86 = 26.14$$

$$u_2 = 26.14 / 2.16 = \mathbf{12.09 \text{ m/s}}$$

Final Answer: The velocity with which he threw the second stone is 12.09 m/s.

12. Study the given figure and answer the following questions: (a) What type of transformer is shown? (b) Why is this used in power station?

(a) Type of transformer: Step-down Transformer (because the secondary coil has fewer wire loops than the primary coil).

(b) Purpose in power grid: It is used to lower the high transmission voltage down to a safe level (like 220V) before electricity enters our homes.

13. Niva made the device as shown in figure (Electric Motor) but rate of rotation of coil is found to be less. What should be done to increase rate of rotation of coil? Write any two methods.

Two standard methods to speed up the motor rotation are:

1. Increase the strength of the electric current flowing from the battery into the coil.
2. Increase the total number of wire turns wrapped around the moving motor frame.

14. Suppose, equal volume of hydrochloric acid is kept in conical flask A, B and C. If 10 g powder limestone is kept in A and C, and piece of 10 gram limestone is kept in B, in which conical flask rate of chemical reaction is fast? Write with reason.

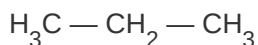
The reaction rate will be fastest in **conical flasks A and C**.

Reason: Powdered limestone has a much larger surface area than solid block pieces. A larger surface area allows more acid molecules to hit the limestone particles at the exact same time, which increases the speed of the chemical reaction.

15. A hydrocarbon compound is a colourless and viscous fluid with sweet taste, used as a sweetening agent. (a) Write its IUPAC name. (b) Write structural formula after hydrogen reaction.

(a) IUPAC name: Glycerol (or Propane-1,2,3-triol).

(b) Saturated structure: Glycerol is already a full saturated compound. Its base three-carbon chain layout is Propane:



16. Calcium carbide fruit ripening text questions: (a) Write an effect seen in human health except above mentioned. (b) Write a process to ripening food by traditional method.

(a) Additional health effect: It contains chemical trace impurities that can cause severe wounds and ulcers inside our throat and digestive food pipe if eaten.

(b) Traditional method: Wrapping the raw green fruits in dry straw packs inside a dark, tight jute bag to trap natural ethylene gas.

Group 'D' — Long Answer Questions (7 x 4 = 28 Marks)

17. Educational technology blog analysis: (a) Which technology? (b) Write two disadvantages. (c) Mention one use of AI.

(a) Technology described: Digital Technology / Computer Technology in Education.

(b) Two simple disadvantages:

1. It can easily distract students from studying via online games and social media apps.
2. Staring at screens for long school hours causes high eye strain and headaches.

(c) One simple use of AI: It acts like an automated personal tutor that changes lessons to match a student's individual learning speed.

18. Newspaper article regarding Dolly the Sheep: (a) Which sheep is Dolly identical to? (b) What is meant by 'a very small piece'? (c) Are the two reasons scientific?

(a) Identical match: Dolly is completely identical to **Sheep 1** (the donor sheep that provided the udder cell nucleus).

(b) Meaning of the phrase: It refers to a microscopic single somatic body cell (an udder cell) containing a full set of DNA.

(c) Scientific evaluation:

- Reason 1 (Cloned people more sensitive to disease): **Yes** (This is a valid biological risk).
- Reason 2 (People should not take the role of a Creator): **No** (This is a moral/religious belief, not a lab fact).

19. Write any two symptoms of high blood pressure and any two preventive measures.

Two common textbook symptoms:

1. Frequent dizzy spells and heavy headaches located at the back of the head.
2. Chest discomfort and sudden pounding heartbeat sensations.

Two simple preventive measures:

1. Reduce the daily amount of common salt eaten in packaged foods and fatty junk snacks.
2. Walk regularly and maintain low stress levels.

20. Given graph shows relation between volume of water and temperature: (a) Name the special property. (b) Write density changes from 0 to 10 degrees. (c) Write an effect.

(a) Name of special property: Anomalous expansion of water.

(b) Density changes process: Density behaves inversely to volume (Density = Mass / Volume). When water is heated from 0°C to 4°C, it shrinks, so its **density increases to reach its maximum peak value at 4°C**. When heated further from 4°C to 10°C, it expands, which causes its **density to decrease steadily**.

(c) Ecological effect: It helps fish and underwater life stay alive during winter. Ice forms only on the lake top because it is light, while the heavy 4°C liquid water settles safely at the bottom pool so aquatic life does not freeze.

21. (a) Write any two differences between concave lens and convex lens. (b) Why is air called a rarer medium and glass called a denser medium?

(a) Two textbook differences:

1. A convex lens is thick in the middle and thin at the borders, while a concave lens is thin in the middle and thick at the borders.
2. A convex lens brings parallel light rays together (converges), while a concave lens spreads light rays apart (diverges).

(b) Reason for medium density names: Air is called an optically rarer medium because light travels very fast inside it (3×10^8 m/s). Glass is called a denser medium because its packed particles slow the light down significantly (2×10^8 m/s).

22. Electronic configurations: A = $1s^2, 2s^2 2p^6, 3s^2 3p^6, 4s^1$ and B = $1s^2, 2s^2 2p^4$. (a) Group of A? (b) Period of B? (c) Molecular formula? (d) Why is B in p-block?

(a) Group of element A: Group IA (or Group 1), because it has 1 valence electron in its outer shell ($4s^1$).

(b) Period of element B: Period 2, because its electrons fill up to a maximum of 2 main energy shells.

(c) Molecular formula: A_2B (Valency of A is 1, Valency of B is 2. Criss-crossing them gives A_2B).

(d) Reason for p-block placement: Element B is kept in the p-block because its last outer electron fills into a 'p' subshell path ($2p^4$).

23. Describe laboratory preparation of ammonia gas and write down a method to test this gas in the gas jar.

Textbook Lab Setup: Ammonia gas is prepared by heating a dry mixture of Ammonium Chloride (NH_4Cl) and Slaked Lime ($Ca(OH)_2$) inside a hard glass test tube.

Method to test the gas jar: Bring a piece of damp (moist) **red litmus paper** near the mouth of the upside-down gas jar. Because ammonia is a basic/alkaline gas, it will immediately turn the red paper **blue**.